Name: _____

WORKSHEET

Derivatives of linear products

1 Find the derivatives of the following:

a $3x^5$ _____

b $2x^6$ _____

c $7x^2$ _____

d $5x^4$ _____

e $-9x^3$ _____

f $-6x^{10}$ _____

g $-2x^5$ _____

h $-7x^3$ _____

2 Find the following

a $\frac{d}{dx}(5x^7)$ _____

b $\frac{d}{dx}(3x)$ _____

c $\frac{d}{dx}(3x^7)$ _____

d $\frac{d}{dx}(-x^3)$ _____

e $\frac{d}{dt}(8t^5)$ _____

f $\frac{d}{dz}(-6z^4)$ _____

g $\frac{d}{dp}(0.5p^8)$ _____

h $\frac{d}{dm}\left(\frac{2}{3}m^9\right)$ _____

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i $\frac{d}{dx}(cx^6)$ _____

j $\frac{d}{dx}(ax^n)$ _____

3 Find the derivatives of the following

a $8x^3 + 2x^4$ _____

b $3x^3 - 7x^4$ _____

c $8x - 5x^2$ _____

d $-6x^5 - 3x^4$ _____

e $-7 + 2x^2$ _____

f $9x^{14} - 17x$ _____

g $12x^5 - \frac{1}{3}x^3$ _____

h $\frac{1}{2}x^3 - \frac{3}{5}x^2$ _____

4 If $f'(x) = 4x^3 - 5x^2$, find:

a $\frac{d}{dx}[2f(x)]$ _____

b $\frac{d}{dx}[-3f(x)]$ _____

c $\frac{d}{dx}[\frac{1}{2}f(x)]$ _____

d $\frac{d}{dx}[-4f(x)]$ _____

5 If $f(x) = 5x^3 - 4x^2$, find the derivatives of the following:

a $f(x)$ _____

b $2f(x)$ _____

c $-4f(x)$ _____

d $\frac{2}{3}f(x)$ _____

e $-\frac{1}{2}f(x)$ _____

f $af(x)$ _____

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Answers

1 **a** $15x^4$

b $12x^5$

c $14x$

d $20x^3$

e $-27x^2$

f $-60x^9$

g $-10x^4$

h $-21x^2$

2 **a** $35x^6$

b 3

c $21x^6$

d $-3x^2$

e $40t^4$

f $-24z^3$

g $4p^7$

h $6m^8$

i $6cx^5$

j nax^{n-1}

3 **a** $24x^2 + 8x^3$

b $9x^2 - 28x^3$

c $8 - 10x$

d $-30x^4 - 12x^3$

e $4x$

f $36x^3 - 17$

g $60x^4 - x^2$

h $\frac{3}{2}x^2 - \frac{6}{5}x$

4 **a** $8x^3 - 10x^2$

b $-12x^3 + 15x^2$

c $2x^3 - \frac{5}{2}x^2$

d $-16x^3 + 20x^2$

5 **a** $15x^2 - 8x$

b $30x^2 - 16x$

c $-60x^2 + 32x$

d $10x^2 - \frac{16}{3}x$

e $-\frac{15}{2}x^2 + 4x$

f $15ax^2 - 8ax$

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